CS 3700 - Final Project

Business-Driven Database Design & Implementation

Fall 2026

Overview

For this project, you will develop a relational database for a business or organization of your choice. You are responsible for defining the business requirements, designing and normalizing the database, implementing it using SQL, and demonstrating your work.

Your final submission must include a **written report, working SQL database, and presentation with a live demonstration**.

This project evaluates your ability to independently apply database concepts, explain your decisions, and solve problems using SQL.

Project Requirements

1. Business Scenario & Requirements

Choose a realistic business or organizational problem that requires a relational database.

Include:

- Business description and problem statement
- At least 3 stakeholders or user types
- At least 5 business rules
- At least 5 functional requirements
- Data requirements
- At least 2 operational challenges specific to your business

Individual Requirement: Each student must receive approval for their business proposal. The instructor will assign one additional business requirement that must be incorporated into the project.

Generic business scenarios without meaningful operational details may be returned for revision.

2. ER Diagram & Database Design

Create an ER diagram using Crow's Foot notation.

Requirements:

- At least 5–7 entities
- Attributes and primary keys
- Relationships and cardinalities
- At least one many-to-many relationship
- Associative entities where necessary

Explain at least 3 major design decisions.

Submit both your initial and revised ER diagrams, including a brief explanation of the changes.

3. Normalization

Demonstrate normalization from **UNF → 1NF → 2NF → 3NF**.

Include:

- An initial unnormalized dataset with at least 10 sample records
- Each normalization stage and resulting tables
- Relevant functional dependencies
- Partial and transitive dependencies, where applicable
- At least 2 data anomalies and how normalization addresses them

Explain why each decomposition is necessary.

In-Class Verification: The instructor may select a table and ask you to explain its normalization or modify its structure.

4. Relational Schema

Document your final database tables, including:

- Column names and data types
- Primary and foreign keys
- Appropriate constraints
- Relationships between tables

Justify at least 5 data type selections and explain at least 3 database constraints.

5. SQL Implementation

Implement your database using SQL.

Requirements:

- `CREATE TABLE` statements
- Primary and foreign key constraints
- `NOT NULL` and `UNIQUE` where appropriate
- At least 5–10 realistic records per table, where appropriate
- At least 3 business edge cases

Demonstrate at least 3 database tests, including one operation intentionally rejected by a constraint.

6. SQL Queries

Write at least **8 meaningful SQL statements** related to your business.

Your SQL must collectively demonstrate:

- `SELECT`, `WHERE`, and `ORDER BY`
- `INNER JOIN` and `LEFT JOIN`
- `INSERT`, `UPDATE`, and `DELETE`
- Queries involving multiple tables

For each statement, provide:

- Business purpose
- SQL statement
- Execution results

Select 2 JOIN queries and explain why the tables and JOIN conditions are appropriate.

Written Report

Submit a **formal report (PDF, approximately 8–12 pages)** covering:

1. **Introduction:** Business background and problem
2. **Business Requirements:** Stakeholders, business rules, and functional requirements
3. **ER Design:** Diagram, relationships, and design decisions
4. **Normalization:** UNF through 3NF with explanations

5. **Relational Schema:** Tables, keys, data types, and constraints
6. **SQL Implementation:** Database creation, sample data, and testing
7. **SQL Queries:** Statements, results, and business relevance
8. **Design Revisions:** At least 2 meaningful changes and their justification
9. **Conclusion:** How the database solves the business problem

Include diagrams, screenshots, and supporting evidence.

Your report must describe your actual database, not a hypothetical implementation.

Presentation & Live Demonstration

Each student will complete an **8–10 minute presentation**, including:

- Business problem and requirements
- ER diagram and normalization
- Major design decisions
- Live database demonstration
- Instructor questions

Live SQL Challenge

During the presentation, the instructor will assign an unannounced SQL task using your database.

You may be asked to:

- Write a new query involving a JOIN and WHERE condition
- Modify an existing query
- Update a record under specified conditions
- Explain an error or constraint violation
- Explain how a business rule is implemented

The challenge must be completed without AI assistance.

Students must be able to explain and defend their submitted work.

Project Checkpoints

Checkpoint	Required Submission
1. Business Proposal	Business problem, rules, functional requirements, and data needs
2. Database Design	ER diagram, normalization, and proposed relational schema
3. Implementation	Working database, sample data, queries, and tests

Checkpoint Verification

Each checkpoint includes a brief instructor review or in-class explanation.

Students may be asked to:

- Explain a design decision
- Identify a normalization problem
- Modify a business rule
- Explain or correct an SQL statement

Checkpoint submissions must be preserved. Changes made after feedback must be reflected in the final project.

Final Deliverables

Submit:

1. **Final Report (.pdf)** — Complete database documentation
2. **SQL File (.sql)** — Table creation, sample data, queries, and tests
3. **Presentation Slides** — Summary of the project
4. **Design Evidence** — Initial and revised diagrams and checkpoint submissions
5. **Live Demonstration** — Database demonstration and SQL challenge

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Grading Rubric (100 Points)

Category	Points
Business Scenario & Requirements	10
ER Diagram & Design Justification	10
Normalization (UNF → 3NF)	15
Relational Schema & Data Types	10
SQL Implementation & Testing	15
SQL Queries & Business Operations	10
Written Report & Documentation	10
Presentation & Live Technical Defense	20
Total	**100**

Presentation Grading (20 Points)

Component	Points
Business and design explanation	5
Working database demonstration	5
Unassisted live SQL challenge	7
Technical questions and answers	3
Total	**20**

Academic Integrity & AI Usage Policy

Permitted AI Use

AI tools may be used to:

- Explain database concepts
- Review general SQL syntax

- Help understand error messages
- Provide general study examples unrelated to your project

Prohibited AI Use

AI tools may **not** be used to:

- Generate your business scenario or requirements
- Create or revise your ER diagram
- Perform your normalization process
- Generate SQL statements for your project
- Write or rewrite your final report
- Generate explanations of your design decisions
- Complete checkpoint verification or live challenges

Academic Integrity

Students are responsible for ensuring that submitted work is their own and complies with the university Academic Integrity and Course policy.

A student who cannot explain or demonstrate a required skill may lose points on the corresponding assessment.

AI disclosure: Any AI assistance must be disclosed, including the tool used and its purpose.

Suspected unauthorized AI use will be evaluated using available evidence and applicable university academic integrity procedures.

Final Note

This project is designed to assess your understanding of database development, not your ability to generate a completed submission.

You must be able to **design, implement, explain, and modify your database independently**.

A functional database is important, but your ability to demonstrate how it works and justify your decisions is equally important.